

Quick Check

Lesson 2: What Is Pressure? Measuring the Invisible Push — Shoes, Scales, and Force Per Unit Area

10 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.1: Pressure | Lesson 2

A plastic jerrycan had some air removed from inside, sealed, and then left on a bench. It has collapsed inwards. Which statement best describes the forces the outside air exerts on this jerrycan?

- A** No air is pushing on it, because the air outside cannot be seen or felt.
- B** The air pushes only on the top of the jerrycan, pressing it downwards.
- C** The air pushes inwards on every face of the jerrycan: top, bottom and all four sides.
- D** The air inside has pushed in on the walls and made the jerrycan collapse.

A teacher presses a flat 300 g textbook on her palm, then presses a sharpened pencil tip on her palm with about the same force. The pencil feels much sharper and more painful. What is the main reason?

- A** The pencil tip touches a much smaller area, so the same force gives a greater pressure.
- B** The pencil tip touches a much smaller area, so it gives a smaller pressure.
- C** The pencil is harder than the book, so its force is larger.
- D** The pencil pushes with a much bigger force than the book.

A student has a mass of 50 kg ($g = 10 \text{ N/kg}$). She stands on one shoe. The shoe tracing on graph paper has an area of 200 cm^2 . Use half her body weight as the force on one foot. What pressure does her shoe exert on the floor?

A 12 500 Pa

B 25 000 Pa

C 1.25 Pa

D 5 Pa

In the shoe investigation, a pair counts the squares in a shoe tracing and finds an area of 250 cm^2 . To calculate pressure in pascals, they must convert this to m^2 . What is the area in m^2 ?

A 0.25 m^2

B 0.025 m^2

C 0.0025 m^2

D 2.5 m^2

Which statement correctly gives the definition of pressure and its SI unit?

- A** Pressure = force $\div$ area, measured in newtons (N)
- B** Pressure = area $\div$ force, measured in m^2/N
- C** Pressure = force $\div$ area, measured in pascals (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$
- D** Pressure = force $\times$ area, measured in $\text{N}\cdot\text{m}^2$

A maize bag of mass 90 kg ($g = 10 \text{ N/kg}$) is placed on a wooden pallet with 4 legs. Each leg has a base area of 25 cm^2 . The weight is shared equally by the legs. What pressure does each leg exert on the warehouse floor in Eldoret?

A 9 Pa

B 360 000 Pa

C 36 Pa

D 90 000 Pa

A Maasai elder leans on a rungu with a handle tip area of 2 cm^2 , using a force of 150 N . What pressure does the rungu tip exert on the ground?

A $750\,000 \text{ Pa}$

B 75 Pa

C $7\,500 \text{ Pa}$

D $30\,000 \text{ Pa}$

A rungu tip (area 2 cm^2) and a flat crutch base (area 50 cm^2) each carry the same force of 150 N . Which is safer on a soft floor, and why?

- A** The crutch, because a larger area gives a smaller pressure on the floor.
- B** The rungu, because it spreads the force over more area.
- C** The rungu, because a small area gives a smaller pressure.
- D** Both are equally safe, because the force is the same.

A student must write a 'What I now THINK' card explaining why a sealed jerrycan with some air removed collapses. Which card is the best scientific explanation using force per unit area?

- A** The air did it.
- B** Air outside pushes on every unit of area of the jerrycan. With less air inside, the outside pressure is greater than the inside pressure, so the walls are pushed in.
- C** Pressure cannot act on plastic, so the plastic became weak by itself.
- D** The air outside got heavier and squashed the jerrycan.

A full jerrycan does not collapse, but the same jerrycan collapses when some air is removed. Which sentence correctly completes the model: 'The full jerrycan stayed intact because ____.'

- A** air pressure pushes only from above, and the plastic is strong enough to resist it
- B** the air pressure inside pushing outwards balanced the air pressure outside pushing inwards
- C** there is no air pressure acting on a full jerrycan
- D** the air outside is lighter than the air inside a full jerrycan

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).